14. Isopropylbenzene on air oxidation in the presence of dilute acid gives

a) C_6H_5COOH b) $C_6H_5COCH_3$ c) $C_6H_5COC_6H_5$ d) C_6H_5 - OH

15. Assertion : Phenol is more reactive than benzene towards electrophilic substitution reaction
Reason : In the case of phenol, the intermediate arenium ion is more stabilized by resonance.

a) if both assertion and reason are true and reason is the correct explanation of assertion.

b) if both assertion and reason are true but reason is not the correct explanation of assertion.

c) assertion is true but reason is false

d) both assertion and reason are false.

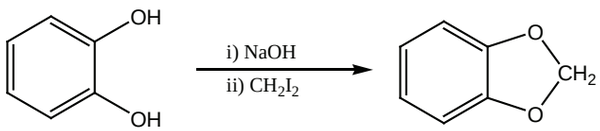
16. HO CH_2 CH_2 – OH on heating with periodic acid gives

a) methanoic acid b) Glyoxal c) methanal d) CO_2

17. Which of the following compound can be used as artifreeze in automobile radiators?

a) methanol b) ethanol c) Neopentyl alcohol d) ethan -1, 2-diol

18. The reactions



is an example of

a) Wurtz reaction b) cyclic reaction c) Williamson reaction d) Kolbe reactions

19. One mole of an organic compound (A) with the formula C_3H_8O reacts completely with two moles of HI to form X and Y. When Y is boiled with aqueous alkali it forms Z. Z answers the iodoform test. The compound (A) is

a) propan – 2-ol b) propan -1-ol c) ethoxy ethane d) methoxy ethane

20. Among the following ethers which one will produce methyl alcohol on treatment with hot HI?

a) $(H_3C)_3C-O-CH_3$ b) $(CH_3)_2-CH-CH_2-O-CH_3$

c) $CH_3(CH_2)_3-O-CH_3$ d) $CH_3-CH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}-O-CH_3$

21. Williamson synthesis of preparing dimethyl ether is a / an /

a) SN^1 reactions b) SN^2 reaction

c) electrophilic addition d) electrophilic substitution

22. On reacting with neutral ferric chloride, phenol gives

a) red colour b) violet colour c) dark green colour d) no colouration.

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